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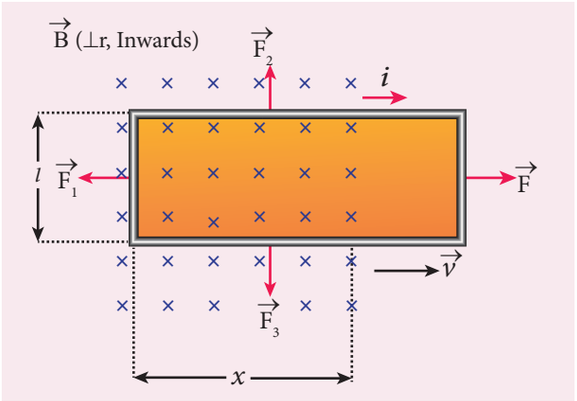


Figure 4.10 Motional emf from Faraday's law

When the loop is pulled with a constant velocity $\vec{v}$ to the right, the area of the portion of the loop within the magnetic field will decrease. Thus, the flux linked with the loop will also decrease. According to Faraday's law, an electric current is induced in the loop which flows in a direction so as to oppose the pull of the loop.

Let x be the length of the loop which is still within the magnetic field, then its area is lx. The magnetic flux linked with the loop is

$$\Phi_B = \int_A \vec{B}\cdot d\vec{A} = BA\cos\theta$$

Here $\theta = 0^\circ$ and $\cos 0^\circ = 1$

$$= BA$$

$$\Phi_B = Blx \qquad (4.8)$$

As this magnetic flux decreases due to the movement of the loop, the magnitude of the induced emf is given by

$$\varepsilon = \frac{d\Phi_B}{dt} = \frac{d}{dt}(Blx)$$

Here, both B and l are constants. Therefore,

$$\varepsilon = Bl\frac{dx}{dt}$$

$$\varepsilon = Blv \qquad (4.9)$$

where $v = \dfrac{dx}{dt}$ is the velocity of the loop. This emf is known as motional emf since it is produced due to the movement of the loop in the magnetic field.

From Lenz's law, it is found that the induced current flows in clockwise direction. If R is the resistance of the loop, then the induced current is given by

$$i = \frac{\varepsilon}{R}$$

$$i = \frac{Blv}{R} \qquad (4.10)$$

Energy conservation

In order to move the loop with a constant velocity $\vec{v}$, a constant force that is equal and opposite to the magnetic force, must be applied. Therefore, mechanical work is done to move the loop. Then the rate of doing work or power is

$$P = \vec{F}\cdot\vec{v} = Fv\cos\theta$$

$$= Fv \qquad \text{Here } \theta = 0^\circ \qquad (4.11)$$

Now, let us find the magnetic force acting on the loop due to its movement in the magnetic field. Let three deflecting forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ be acting on the three segments of the loop as shown in Figure 4.10. The general equation of such a deflecting force is given by

$$\vec{F}_d = i\,\vec{l}\times\vec{B}$$

Forces $\vec{F}_2$ *and* $\vec{F}_3$ are equal in magnitude and opposite in direction and cancel each other. Therefore, the force $\vec{F}_1$ alone acts on the left segment of the loop in a direction shown in Figure 4.10 and is given by

$$\vec{F}_1 = i\vec{l}\times\vec{B}$$

$$F = ilB\sin\theta \qquad (4.12)$$

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